

Preliminary Project Assessment

Project Name: NFL Multi-Platform Trading Dashboard

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1. Executive Summary

This document summarises our understanding of your project and outlines how we plan to approach it based on the requirements shared. Our goal is to ensure that we are fully aligned with your vision, expectations, and the intended outcomes before moving into detailed planning.

Here, we've captured our interpretation of the objectives, the proposed high-level architecture, and the overall approach we believe will best support the project goals.

This is not the final **Scope of Work (SOW)** but rather a foundation for it — a way to confirm that we share the same understanding and direction before we move ahead with the detailed technical and execution plan.

2. Our Assessment of Requirement

The Project requires a **unified trading dashboard** for professional traders who wish to trade NFL sports markets across multiple platforms. The system should:

- Fetch and display **live NFL market data from Pinnacle** (reference feed).
- Allow traders to **Analyse odds** and **Post corresponding quotes** on multiple platforms (e.g., **Kalshi, ProphetX**) via API integrations.
- Support **Manual** as well as **Automated trading**, where automation adjusts quotes and trade behaviour based on Pinnacle's live odds.
- Implement **Risk Management** controls (stop-loss, quote limits, exposure tracking).
- Record all trades and generate **Real-time Profit and Loss (P&L)** summaries per user, market, and platform.
- Maintain **Counterparty tracking** to monitor performance of traders (sharp vs weak bettors) for future quote adjustments.



- Provide **Notifications and Logs** for user alerts, trade confirmations, and system events.

Essentially, the platform acts as a **Smart Bridge** between Pinnacle (reference market) and other trading platforms, automating quote posting and ensuring safe, risk-controlled operations.

3. Developer's Solution and Deliverables

We will have to build an end-to-end system that fulfils the above goals with the following deliverables:

- User Authentication & Access**
 - Secure login via username/password.
 - API key registration for automated Quoting and Quote Posting.
- Live Market Feed Integration (Pinnacle)**
 - Establish WebSocket & REST connection to Pinnacle's live NFL data feed.
 - Cache live odds and market information for rapid internal access.
- Target Platform Connectivity**
 - Integrate APIs for Kalshi and ProphetX and other mentioned Platforms.
 - Enable quote posting, modification, and cancellation directly from the dashboard.
- Trading & Automation Engine**
 - Allow manual quote posting or enable auto-trading based on Pinnacle's odds.
 - Apply user-defined automation rules (spread difference, liquidity conditions, etc.)
 - Log every trade for audit and future reference.
- Risk Management Layer**
 - Enforce user-level and market-level limits (e.g., quote size, position exposure, stop-loss).
 - Dynamic restriction based on trader or counterparty behaviour.
- Profit & Loss Tracking**
 - Real-time P&L per market, per user, per platform.
 - Historical performance analytics.
- Counterparty Tracking**
 - Maintain historical behaviour of bettors interacting with trader quotes.
 - Identify "sharp" or "weak" bettors to optimize quote size and spread width in future trades.



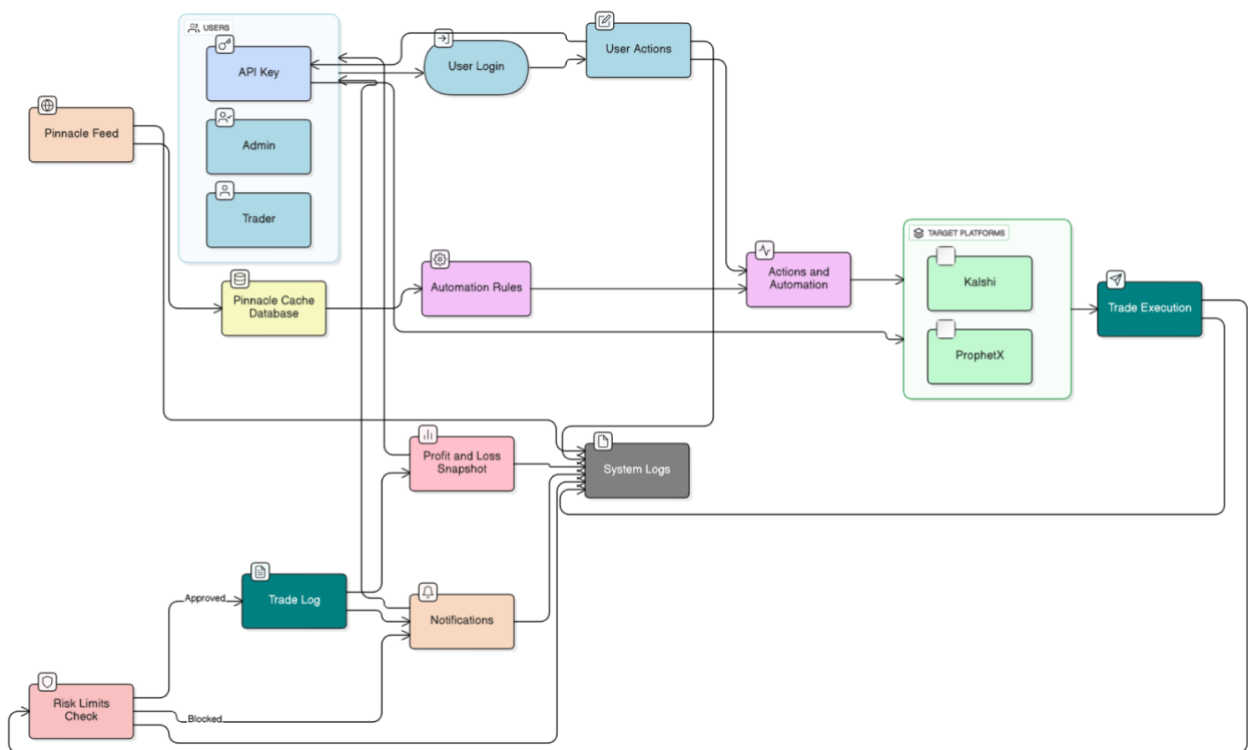
h) Monitoring, Notifications & System Logs

- Real-time alerts for trade executions, errors, and system anomalies.
- Central log system for debugging and audit trail.

i) Web Dashboard Interface

- Intuitive frontend showing markets, odds, automation controls, trades, and alerts.
- Separate views for traders and admins.

4. Data Flow Charts



5. Proposed Tech Stack

Layer	Technology / Tools	Purpose
Frontend	React.js	User dashboard and trading UI
Backend	Python (Fast API)	Core API server handling trading logic
Database	PostgreSQL	Structured storage for users, trades, limits, logs
Caching / Queue	Redis	Pinnacle feed caching and background job handling
Automation / Workers	Celery / Async task	Handle scheduled quote posting and updates
APIs Integration	REST & WebSocket	Connect with Pinnacle, Kalshi, ProphetX
Authentication	JWT / OAuth	Secure login and API token management
Deployment	Docker + Nginx	Containerized, scalable deployment
Monitoring / Logging	Prometheus, Grafana	System health and trade activity visualization

6. Expected Outcome

The final system will allow traders to:

- Seamlessly compare live Pinnacle NFL odds and post optimized quotes to Kalshi and ProphetX.
- Enable automation rules that trade hands-free while respecting defined risk limits.
- Gain analytical insight through P&L, counterparty stats, and historical logs.
- Operate in a secure, scalable, and fully auditable environment.

